

Chromatography of Food Dyes

Exploration

Separations of mixtures can be accomplished using physical properties and changes.

Paper chromatography is based on adsorption of a substance to paper vs its solubility in a solvent.

The more affinity for the solvent, the further the substance moves up the paper.

R_f factor or retention factor

$$\frac{\text{Distance spot moves}}{\text{distance solvent moves}} = R_f$$

A substance will have the same R_f value irrespective of experimental conditions

Experimentation

- No pens! Pencil only
- use a ruler to measure the paper
- use a ruler to measure 1 cm from the long edge and draw a line w/ straight edge
- when you dot the paper, smaller dots are better.
- Don't let the edges of the filter paper touch!
(no matter what the directions say!)

1 SQUARE = _____

- Mark the solvent front across the whole paper, not just the edge.

- the solvent front might not be the same across the whole chromatogram